

OPEN SOURCE AI AVATAR PLATFORM

2-Developer Parallel Architecture, Contract-First Engineering & Equal 50/50 Roadmap

Project Model: 2-Developer Vertical Pillars	Architecture: Contract-First / Asynchronous Queues	Target: Real-Time & Batch Avatars
Developer 1: Audio AI, Voice & Core Pipeline	Developer 2: Vision AI, Lip Sync & Interactive UX	Version: 1.0 (Master Blueprint)

1. Strategic Division: Audio/Backend vs. Vision/Interaction Pillars

Building an enterprise-grade AI Avatar Creation Platform requires converging Neural TTS, Zero-Shot Voice Cloning, Diffusion Lip Synchronization, 3D Facial Rigging, and Real-Time WebRTC Streaming. Rather than assigning one engineer to frontend and the other to backend, work is divided into **two vertical full-stack engineering domains** bounded by canonical API contracts.

DEVELOPER 1 Audio AI, Voice Synthesis & Backend Core

Primary Mandate: Owns the neural voice synthesis engine, voice cloning, audio DSP, task queue orchestration, and backend infrastructure.

- **Tech Stack:** Python / PyTorch, FastAPI, Celery, Redis, PostgreSQL, Docker/K8s
- Multi-Model Neural TTS (Higgs Audio V2 5.77B, Dia, Kokoro v1.0, MMS-TTS)
- Zero-Shot Voice Cloning & Cross-Lingual Style Transfer (OpenVoice V2, XTTS-v2)
- Phoneme/Viseme Alignment Extraction & Audio Feature Normalization
- Distributed Task Queue, GPU Worker Dispatch & Media Storage (S3/GCS)
- Audio Ethical Safeguards (Biometric Voice Consent & Audio Watermarking)

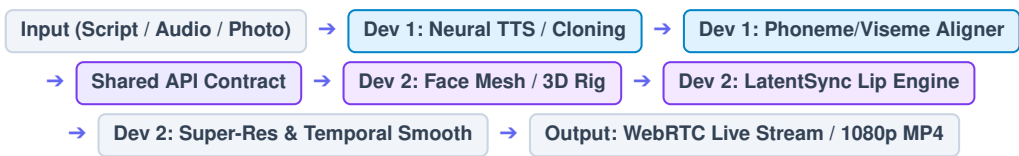
DEVELOPER 2 Vision AI, Lip Sync, Avatar Synthesis & UI Core

Primary Mandate: Owns facial animation, deep learning lip sync, procedural avatar generation, super-resolution, and real-time interactive UI.

- **Tech Stack:** PyTorch, LatentSync / MuseTalk, MediaPipe / FLAME, React / Electron, WebRTC
- High-Fidelity Lip Synchronization (>95% precision via LatentSync / MuseTalk)
- Photo-to-Avatar & Procedural Face Generation (Stable Diffusion / StyleGAN3)
- Temporal Consistency Optimization & Super-Resolution (Real-ESRGAN)
- WebRTC Sub-200ms Real-Time Streaming & Interactive Avatar Canvas
- Visual Ethical Safeguards (Invisible Watermarking, C2PA Provenance, Deepfake Shield)

2. Production Avatar Pipeline & Shared Contracts

The system executes a unified production flow where audio features and visual rendering pipelines interface through timestamped, phoneme-aligned data contracts.



Shared Audio-Visual Alignment Contract (POST /api/v1/avatar/render-job):

```
{
  "jobId": "AVT-9821-X",
  "avatarId": "AVATAR_FEMALE_04",
  "audioUrl": "s3://assets/audio/generated_tts_9821.wav",
  "sampleRate": 24000,
  "durationSeconds": 14.5,
  "phonemeTimestamps": [
    {"phoneme": "HH", "viseme": "viseme_sil", "startMs": 0, "endMs": 120},
    {"phoneme": "EH", "viseme": "viseme_E", "startMs": 120, "endMs": 280},
    {"phoneme": "L", "viseme": "viseme_L", "startMs": 280, "endMs": 410},
    {"phoneme": "OW", "viseme": "viseme_O", "startMs": 410, "endMs": 620}
  ],
  "emotionVector": {"happy": 0.8, "neutral": 0.2, "eyeblinkRate": 1.2},
  "renderQuality": "1080P_HQ",
  "targetFps": 30
}
```

3. Phase-by-Phase Roadmap & Integration Gates

PHASE 0 Infrastructure Setup, GPU Environment & Contract Freeze (Days 1–3)

Developer 1 (Backend & Model Hub)

- Scaffold FastAPI service with Redis, Celery & PostgreSQL.
- Configure model hub storage, HuggingFace caching & CUDA runtime.
- Define OpenAPI contract for audio synthesis and avatar render jobs.

Developer 2 (Vision Core & Frontend Shell)

- Scaffold React/Electron Creator Studio with live canvas viewport.
- Configure PyTorch / CUDA vision environment (LatentSync, OpenCV).
- Build mock render client using mock phoneme timestamp payloads.

Integration Gate 0: Contracts frozen. Mock audio/viseme payloads render dummy avatar bounding boxes successfully.

PHASE 1 Core TTS Engine vs. Face Detection & Landmark Rigging

Developer 1 (Neural TTS Engine)

- Integrate Kokoro v1.0 (82M) for sub-second low-latency synthesis.
- Integrate Higgs Audio V2 (5.77B) & Dia for ultra-high MOS (>4.0) speech.
- Build automated model router based on language, speed & style requests.

Developer 2 (Facial Analysis & Rigging)

- Implement MediaPipe Face Mesh (468 3D facial landmarks).
- Build FLAME 3D face mesh fitting & head pose estimator.
- Create facial bounding box cropper and background segmenter.

Integration Gate 1: Text input generates speech audio; face detector accurately maps 468 landmarks on reference photos.

PHASE 2 Zero-Shot Voice Cloning vs. LatentSync Lip Synchronization

Developer 1 (Voice Cloning & Alignment)

- Integrate OpenVoice V2 & XTTS-v2 for 30–60s reference audio cloning.
- Implement cross-lingual accent, rhythm & intonation control.
- Build forced-aligner to output millisecond-accurate phoneme timestamps.

Developer 2 (Lip Sync & Viseme Engine)

- Integrate LatentSync / MuseTalk diffusion lip synchronization models.
- Implement cross-lingual phoneme-to-viseme mapping for 20+ languages.
- Achieve >95% frame-level audio-to-lip sync accuracy.

Integration Gate 2 (First Complete Talking Avatar): Audio from cloned voice + Single photo → Accurately lip-synced video clip.

PHASE 3 Multilingual Synthesis vs. Multi-Modal Avatar Customization

Developer 1 (Multilingual & Emotion Engine)

- Integrate Facebook MMS-TTS for 1000+ language support.
- Add emotion prosody vectors (joy, anger, sorrow, authority).
- Build speech quality auditor (automated MOS / PESQ prediction).

Developer 2 (Avatar Generator & Style Transfer)

- Photo-to-Avatar conversion with facial identity preservation (>90%).
- Stable Diffusion & StyleGAN3 procedural avatar generator.
- Customization studio UI: clothing, hair, lighting, background swaps.

Integration Gate 3: User creates customized avatar, selects cloned multilingual voice with emotion, and previews rendering.

PHASE 4 GPU Task Queue & Streaming Audio vs. Real-Time WebRTC Rendering

Developer 1 (Streaming Audio & Batch Workers)

- Sub-100ms streaming TTS engine using RealtimeTTS / chunked Kokoro.
- Celery + Redis distributed job queues for high-throughput batching.
- WebSocket audio stream gateway for interactive avatar sessions.

Developer 2 (WebRTC Engine & Live Control)

- WebRTC video/audio streaming pipeline with sub-200ms latency.
- Live interactive avatar animation driven by streaming audio chunks.
- Super-resolution (Real-ESRGAN) & temporal motion smoothing (<2% jitter).

Integration Gate 4 (Live Interactive Avatar): User speaks or streams text $\rightarrow$ Live animated avatar streams back in real time (<200ms).

PHASE 5 Ethical AI, Watermarking, Provenance & Deepfake Shield

Developer 1 (Voice Consent & Audio Integrity)

- Biometric voice consent verification protocol for voice cloning.
- Inaudible acoustic watermarking for synthetic audio provenance.
- Abuse detection, voice similarity anomaly alerts & rate limiting.

Developer 2 (Visual Watermarking & C2PA Provenance)

- Invisible digital visual watermarking & C2PA metadata injection.
- Deepfake detection harness & authenticity validation badges.
- Ethical disclaimer rendering & consent audit trail logs.

Integration Gate 5: Generated avatar video contains cryptographically verifiable watermarking and passes deepfake safety compliance.

PHASE 6 Production Hardening, Scaling, Testing & Final Showcase

Developer 1 (Backend Reliability & Scale)

- GPU auto-scaling on Kubernetes, AWS P4/P5 or GCP TPU clusters.
- Concurrency stress testing (50+ simultaneous avatar generation tasks).
- Public REST/WebSocket API documentation & Python SDK.

Developer 2 (Video Quality & Frontend Polish)

- Visual fidelity benchmark (LPIPS < 0.1, FID optimization).
- Cross-browser WebRTC testing & edge latency optimization.
- Complete demo showcase: Education, Corporate, Support, Content Avatars.

Final Gate: >99.5% uptime, 50+ concurrent tasks, <30s generation time for 60s video, <200ms real-time latency, >95% lip sync accuracy.

4. Master Timeline & 50/50 Workload Allocation

Phase	Developer 1 (Audio AI & Backend)	Developer 2 (Vision AI & Interactive UX)	Milestone / Integration Gate
Phase 0	FastAPI, Redis, Celery, HuggingFace Hub, API Specs	React Studio Shell, PyTorch CUDA Vision, Mock client	OpenAPI & Alignment Contract Frozen
Phase 1	Kokoro v1.0, Higgs Audio V2, Model Router	MediaPipe 468 landmarks, FLAME 3D, Face segmenter	TTS generated + Facial landmarks fitted
Phase 2	OpenVoice V2, XTTS-v2 Voice Cloning, Phoneme Aligner	LatentSync / MuseTalk Lip Sync (>95% accuracy)	First End-to-End Talking Avatar Video
Phase 3	MMS-TTS (1000+ languages), Emotion Prosody control	Stable Diffusion / StyleGAN3 Avatar Customization Studio	Custom multilingual avatar with emotion
Phase 4	RealtimeTTS streaming, Celery batch worker pools	WebRTC sub-200ms video stream, Real-ESRGAN	Live Interactive Streaming Avatar (<200ms)
Phase 5	Biometric Voice Consent & Audio Watermarking	C2PA Provenance, Visual Watermark, Deepfake Shield	Ethical Compliance & Verification passed
Phase 6	Kubernetes GPU autoscaling, 50+ concurrent stress test	LPIPS/FID benchmark, 4 Showcase Production Demos	Full Production Open Source Release

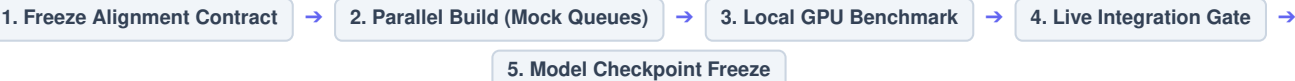
Developer 1 Workload Breakdown (50%)

Neural TTS Multi-Model Engine (Kokoro, Higgs, MMS)	10%
Zero-Shot Voice Cloning & Style Transfer (OpenVoice, XTTS)	10%
Phoneme-to-Viseme Timestamp Aligner & Audio DSP	8%
Distributed Celery/Redis GPU Queue & Backend API	12%
Voice Consent Verification & Audio Watermarking	5%
Backend Load Balancing, GPU Auto-Scaling & SDK	5%
Total Workload Allocation	50%

Developer 2 Workload Breakdown (50%)

LatentSync & MuseTalk Lip Sync Diffusion Engine	12%
MediaPipe 468 Landmarks & FLAME 3D Face Rigging	8%
Avatar Synthesis (StyleGAN3/SD) & Real-ESRGAN Upscaler	10%
WebRTC Real-Time Streaming Sub-200ms Engine	8%
React/Electron Creator Studio & Customization Canvas	7%
C2PA Provenance, Visual Watermarking & Deepfake Shield	5%
Total Workload Allocation	50%

5. Integration Governance & Quality Benchmarks



Quality Benchmark	Target Metric	Verification Method
Lip Sync Accuracy	>95% precision across 20+ languages	LSE-C / LSE-D confidence scores & manual sync audit
Voice Cloning Similarity	>90% similarity from 30–60s reference audio	Speaker verification cosine distance (ECAPA-TDNN)
Batch Video Generation	<30 seconds for 60-second 1080p video	End-to-end GPU benchmark on single NVIDIA RTX 4090 / A100
Interactive Real-Time Latency	<200ms end-to-end streaming latency	WebRTC round-trip timestamp measurement
Temporal Consistency	<2% frame-to-frame feature jitter	Optical flow landmark variance & LPIPS perceptual loss